Journal of Power Sources 435 (2019) 226777

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

Multi-scale thermal stability study of commercial lithium-ion batteries as a function of cathode chemistry and state-of-charge

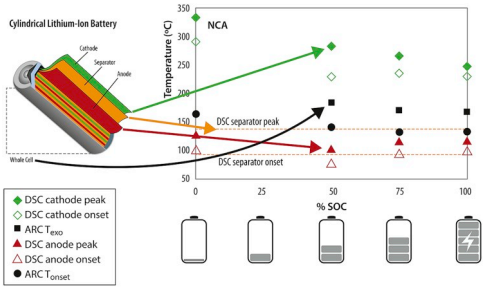
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HIGHLIGHTS

- Thermal runaway measurements for commercial Li-ion batteries at multiple SOC.
- Lithium ion cell component level degradation onset mapped to whole cell runaway.
- Component level degradation from calorimetry related to phase behavior with TR-XRD.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Calorimetry
Thermal runaway
Lithium-ion
Thermogravimetric analysis
Temperature-resolved X-ray diffraction

ABSTRACT

This paper takes a critical look at the materials aspects of thermal runaway of lithium-ion batteries and correlates contributions from individual cell components to thermal runaway trends.

An accelerating rate calorimeter (ARC) was used to evaluate commercial lithium-ion cells based on $LiCoO_2$ (LCO), $LiFePO_4$ (LFP), and $LiNi_xCo_yAl_{1-x-y}O_2$ (NCA) at various states of charge (SOC). Cells were disassembled and the component properties were evaluated by thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), and temperature-resolved X-ray diffraction (TR-XRD).

The whole cell thermal runaway onset temperature decreases and peak heating rate increases with SOC due to cathode destabilization. LCO and NCA cathodes are metastable, with NCA cells exhibiting the highest thermal runaway rates. By contrast, the LFP cathode is stable to >500 °C, even when charged. For anodes, the decomposition and whole cell self-heating onset temperature is generally independent of SOC. DSC exotherm onset temperatures of the anodes were generally within 10 °C of the onset of self-heating in whole cell ARC. However, onset temperatures of the cathodes were typically observed above the ARC onset of whole cell runaway. This systematic evaluation of component to whole cell degradation provides a scientific basis for future thermal modeling and design of safer cells.

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https://doi.org/10.1016/j.jpowsour.2019.226777
Received 25 April 2019; Received in revised form 9 June 2019; Accepted 18 June 2019
Available online 2 July 2019
0378-7753/© 2019 Published by Elsevier B.V.

1. Introduction

High cell voltage and energy density, long lifetime, and reliability have made rechargeable lithium-ion batteries (LIB) the technology of choice for consumer electronics and electric vehicles. They are beginning to find widespread use in grid-tied stationary storage systems [1,2]. However, with increasing energy and power density come heightened risks of thermal runaway or a serious fire [3–10]. Incidents of smoke or fire caused by abuses or failure under normal conditions are still an unresolved hurdle to widespread adoption of lithium-ion batteries in grid energy storage applications. A variety of parameters such as cell chemistry and state-of-charge (SOC) can impact thermal runaway characteristics. For the successful implementation and commercial operation of utility class energy storage plants, component level thermal stability as a function of chemistry and SOC should be understood and related to whole cell phenomena. Understanding cell-level phenomena is a critical foundation for larger-scale thermal failure propagation studies.

The lithium battery industry mainly uses layered transition metal oxides such as $LiCoO_2$ (LCO) and $LiNi_xCo_yAl_{1-x-y}O_2$ (NCA) as cathode materials where power and performance are a key requirement, and $LiFePO_4$ (LFP), an olivine phosphate, where stability and long cycle life are of paramount importance. On the anode side, graphite or other forms of carbon are the most common materials. There have been numerous reports on the thermal stability of various cell components by thermogravimetric analysis (TGA) [11–13], X-ray diffraction (XRD) [14–18], and calorimetry [19–26]. However, these studies may not represent cycled commercial LIB materials as they were not harvested from commercial cells. Additionally, testing conditions and techniques vary by group and publication, making comparative studies across the literature challenging. Of the few studies available on commercial LIBs, most are limited to whole cell accelerated rate calorimetry (ARC) data without component analysis [27–29]. Recent studies of commercial cells incorporated component level characterization, but not across a broad range of SOC or using consistent cell geometries [30,31]. A deeper understanding of commercial cell thermal runaway is required to develop failure mitigation strategies and to design inherently safe energy storage. As such, a multi-scale systematic study of commercial LIBs of several chemistries and SOCs is important. In this work, we employed various methods to investigate the correlation between intrinsic properties of cell components and the thermal stability of whole commercial cells under similar test conditions. Three popular commercial 18650 Li-ion cells with LFP, LCO, and NCA cathodes were purchased from well-known manufacturers. The cells were studied at the whole cell and component level. Relationships between component chemistry, thermal stability, SOC, and thermal runaway behavior will be outlined. By better understanding thermal runaway in commercial cells under systematically varied conditions, the cascade and outcomes of thermal events may be better predicted. This may facilitate a more data-informed development of LIBs and systems that are lower risk, and the development and adoption of more robust failure mitigation strategies.

2. Experimental

2.1. Cell and electrode preparation

The 18650 format LIB were commercially sourced: LCO was purchased from LG Chem, LFP was obtained from A123 Systems, and NCA was sourced from Panasonic (Table S1). The SOC was defined by dividing charging capacity by the average capacity at 100% SOC from three charge/discharge cycles of a pristine cell at C/4 rate at room temperature (a rate of 1C corresponds to the current that will discharge the full capacity of a battery in 1 h). The cell was then charged at C/4 to the desired capacity. 100% SOC is defined as the capacity obtained with a 0.5 A constant current charge with 10% charge rate taper to the maximum manufacturer-specified charging voltage.

Cell disassembly took place within an Ar-filled glove box. The external label and metal 18650 can were removed to yield the electrode jellyroll. Anode, cathode, and separator were parted and samples of each were reserved for future characterization. The manufacturer specifications for the cells indicate an organic electrolyte, which is expected to be a mixture of ethylene carbonate and dimethyl carbonate with $LiPF_6$ salt. In one instance, a disassembled electrode was washed with dimethyl carbonate prior to analysis.

2.2. Accelerating rate calorimetry

ARC testing was performed on the 18650 cells (LFP, LCO, and NCA) at 0, 50, 75, and 100% SOC. The ARC was operated in heat-wait-seek mode to determine the onset of self-heating. In this mode, the ARC increases temperature in discrete steps, waits for the thermal transients to decay, and then monitors the temperature of the cell for a fixed time. If the cell temperature does not increase above a threshold value, the ARC repeats heat-wait-seek. If an exotherm is detected, the ARC measures temperature in an adiabatic state. Once a threshold temperature is reached, the test is ended and the cell cools. ARC experiments represent a thermal abuse environment that includes moderately high temperatures for long periods of time.

ARC data was acquired for LCO at 0% SOC and LFP cells at all SOC using a Thermal Hazard Technology ES-ARC (manufacturer-designated abbreviation). Prior to measurements, a high-pressure test at $>$ 800 psi was performed to verify that the system is leak-tight. Plastic insulating labels were removed from 18650 cells prior to being loaded in the ARC. Cells were confined within a snug metal cell holder attached to a pressure transducer. The bomb thermocouple was attached with a copper mesh to the side of the cell holder in two positions. Immediately prior to measurement an additional low-pressure test to ~50 psi was performed to once again ensure the system is leak-tight. ARC testing was performed from 30 to 450 °C with 0.020 °C min^{-1} slope sensitivity. In the heat-wait-seek procedure, heating was performed in 5 °C steps with a 30-min wait time. For ARC calculations, the sample heat capacity is defined as 0.95 J g^{-1} K^{-1} (masses given in Table S1) and the test cell heat capacity is 0.5 J g^{-1} K^{-1} (mass of 106 g).

ARC data was acquired for the more energetic LCO and NCA cells using a Thermal Hazard Technology Expanded Volume ARC (EV-ARC). Testing was performed from 50 to 405 °C with a 0.020 °C min^{-1} slope sensitivity. Heating was performed in 5 °C steps with a 30-min wait period. 5 °C steps offer a convenient time-scale at which to carry out ARC experiments while not missing the onset of key thermal events. The 30-min wait period is intended to provide enough time for the temperature to reach a steady state and to determine if there is self-heating. For calculations, the sample heat capacity is again defined as 0.95 J g^{-1} K^{-1}, but no test cell holder is used in the EV-ARC. Fig. S1 offers a comparison of the two ARCs.

2.3. Thermogravimetric analysis/differential scanning calorimetry

Simultaneous thermogravimetric analysis/differential scanning calorimetry (TGA/DSC) allows for simultaneous quantification of mass change (TGA) and the corresponding enthalpy response (DSC) to changes in temperature. TGA/DSC is used to evaluate materials decomposition in response to thermal abuse conditions. The data was acquired on the simultaneous TGA/DSC instrument Netzsch STA 449 F1 under N_2 flow (150 mL min^{-1}) at a heating rate of 15 °C min^{-1} (unless specified otherwise). Samples were prepared in the glove box by placing approximately 10–15 mg of powder removed from the current collector, without any drying or washing, into an alumina crucible with a lid. The crucibles were not sealed to avoid pressurization during measurement. The crucible was then placed in an Ar-filled container and transferred out of the glove box. Crucibles were removed from the container immediately prior to the TGA/DSC measurement. The DSC data were not corrected for any baseline shift. The onset temperature was

determined by the automatic procedure in the Netzsch Proteus Thermal Analysis v6.1 package at the intersection of the extrapolated baseline and the tangent line passing through the inflection point on the face side of the exotherm peak. Due to the sharpness of the peaks, the application of more rigorous baseline correction would likely not change the onset.

2.4. Temperature-resolved X-ray diffraction

Temperature-resolved X-ray diffraction (TR-XRD) was carried out on a Rigaku SmartLab II XRD with a HT 1500 high temperature attachment in Parallel Beam Configuration with a graphite monochromator. Electrode materials were scraped off the current collector and powders were placed on an alumina substrate and heated at 15 °C min^{-1} under ~10^{-2} Torr vacuum to a specified temperature. X-ray diffractograms were recorded after a 10-min temperature stabilization period for each temperature point. This process was repeated until all temperature points were sampled (in the range of 25–200 °C for anode materials and 25–600 °C for cathode materials). Electrode samples were collected from the LFP, LCO, and NCA cells at 0 and 100% SOC. The samples were kept in an inert environment until being placed into the high temperature chamber and evacuated.

Unlike the other samples, the LFP cathode powder was analyzed on a zero-background silicon substrate to avoid overlap between Al_2O_3 peaks in the previous substrate and possible Fe_2O_3 peaks in the cathode material. The sample was heated at a rate of 20 °C min^{-1} followed by a 30-min equilibration time before each measurement. The equilibration time was longer for this sample given the greater gap between the temperatures at which each measurement was completed.

3. Results and discussion

For the cathode chemistries studied, the operating voltage is highest for NCA, slightly lower for LCO, and notably lower for LFP at all SOC. The typical voltage profiles of LCO, LFP, and NCA cells as a function of SOC are shown in Fig. S2. Note that both NCA and LCO have an SOC-dependent rise in the voltage with no plateau, while LFP has a very different profile, with a very wide plateau from 10 to 90% SOC.

Whole cell failure from the three cell chemistries using ARC shows a range of heat release rates of five orders of magnitude across chemistries. For each chemistry, ARC data was taken at 0, 50, 75, and 100% SOC. Semi-log plots of the ARC self-heating rate profiles are shown as a function of temperature and SOC in Fig. 1. (The same information is also displayed with linear axes in Fig. S3.) Qualitatively, a parallel is seen between how flat the plateau, or small the voltage change, is in the normal voltage profile from Fig. S2 and how small the variation in heating rate is as a function of SOC within a given chemistry. A cell's first detectable exothermic event (>0.02 °C min^{-1} rate) is given as the onset temperature of self-heating (T_{onset}) in Table 1. Due to the extreme temperatures of some cells' thermal runaway behavior, two different ARCs were used. The ES-ARC was used for lower energy cells (LFPs and LCO at low SOC) and captured pressure data simultaneously. The EV-ARC does not capture pressure data and was used for more energetic failures (LCOs at high SOC and NCAs). Cathode decay energetics can be estimated from the maximum heating rate in the ARC. Therefore, the maximum heating rate (HR_{max}), nominal capacity normalized maximum heating rate ($HR_{max,nominal}$), fractional capacity normalized maximum heating rate ($HR_{max,fraction}$), and vented gas pressure (P_{max}), when available, are recorded in Table 1. $HR_{max,nominal}$ was found using the nominal capacities of each cell (2.5 Ah for LCO, 1.1 Ah for LFP, and 2.9 Ah for NCA). $HR_{max,fraction}$ was found by multiplying $HR_{max,nominal}$ by the fractional SOC. Across chemistries, the $HR_{max,nominal}$ at 100% SOC varies by four orders of magnitude, with 4.1 °C min^{-1} Ah^{-1} in LFP and 7410 °C min^{-1} Ah^{-1} in NCA. For NCA, the $HR_{max,nominal}$ varies by four orders of magnitude from 0% to 100% SOC. LCO varies by three orders of magnitude as a function of SOC, while LFP varies by a single order of magnitude in $HR_{max,nominal}$. These differences in SOC dependence are significant.

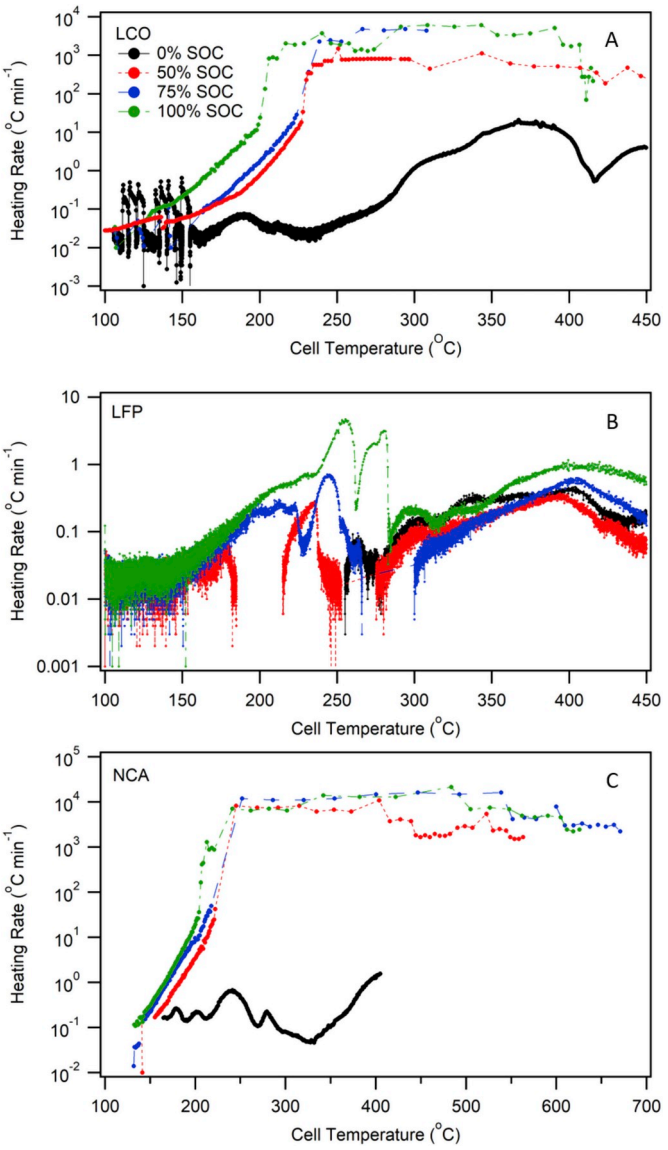


Fig. 1. Log linear plot of heating rate as a function of temperature based on ARC results from (A) LCO, (B) LFP, and (C) NCA cells at 0% SOC (black, solid line), 50% SOC (red, dotted line), 75% SOC (blue, dashed line), and 100% SOC (green, dash-dot line). Circles are data points and lines are a guide to the eye. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

To describe the thermal runaway process, two main steps are identified [32,33]:

(1) Decomposition of the solid-electrolyte interface (SEI) together with the reaction between intercalated lithium and the electrolyte (delithiation), and
(2) Reaction between oxygen from the decomposing cathode and electrolyte

Thermal runaway is initiated via the breakdown of the SEI layer as the cell overheats [7]. The SEI layer decomposes at relatively low temperatures, 60–130 °C, and this process is expected to manifest itself in the ARC as a self-heating event [26]. Once the SEI is decomposed, the electrolyte will exothermically react with exposed lithium embedded in the carbon anode. As the anode decomposition proceeds, hydrocarbon gases are released, building pressure inside the 18650 can. Around

Table 1
Key findings from ARC measurements of thermal runaway behavior as a function of lithium-ion battery chemistry and SOC.

Cell	SOC (%)	ARC type[a]	T_{onset}[b] (°C)	T_{exo}[c] (°C)	HR_{max} (°C min^{-1})	$HR_{max,nominal}$ (°C min^{-1} Ah^{-1})	$HR_{max,fraction}$ (°C min^{-1} Ah^{-1})	P_{max} (psi)
LCO	0	ES	105.8	298.1	18.4	7.4	–	1827.9
	50	ES	96.6	202.5	1116.9	447	894	2981.5
	75	EV	106.5	193.3	4822.3	1930	2570	–
	100	EV	106.4	168.4	6053.5	2420	2420	–
LFP	0	ES	105.9	–	0.4	0.4	–	2900.5
	50	ES	100.8	–	0.3	0.3	0.5	3041.2
	75	ES	100.7	–	0.7	0.6	0.8	3331.9
	100	ES	100.9	240.5	4.5	4.1	4.1	2912.8
NCA	0	EV	164.9	391.8	1.5	0.5	–	–
	50	EV	140.7	182.2	10810.3	3730	7460	–
	75	EV	131.7	169.4	16107.4	5550	7410	–
	100	EV	132.5	165.9	21480.4	7410	7410	–

[a] An ES-ARC (manufacturer-designated abbreviation) is used for less energetic cells, has a larger adiabatic window, and can capture pressure data. An EV-ARC (extended volume) is used for more energetic cells. Images of the two ARCs are provided in Fig. S1.
[b] First detectable exothermic event/onset of self-heating (>0.02 °C min^{-1} change).
[c] Heating rate first surpasses 1 °C min^{-1}.

130–170 °C, the cathode-anode separators start to melt, Fig. 2, allowing the electrodes to short circuit [34]. The ARC data will demonstrate one of two patterns: either a hard (low impedance) internal short circuit from dramatic separator failure, or no hard short occurs and the ARC data would be consistent with thermochemical reaction within the cell coupled with a developing short circuit from a gradually decomposing separator. A gap between the DSC endotherm peak temperature of the separator and the temperature at which sudden high rate thermal runaway occurs in the whole cell ARC data would suggest the latter (a combination of thermochemical reaction and short circuit). This was observed for all cells in Fig. 1. The high runaway rates present for LCO and NCA cells exceed the capabilities of the EV-ARC to provide an adiabatic environment and result in a significant amount of noise in the runaway (Fig. S3 A/C), limiting the amount of quantitative information that can be derived from the ARC data. Despite these limitations, the general trends provide useful information for overall evaluation of the cells. To quantify the onset of large exotherms, the temperature was recorded (T_{exo}) when the heating rate surpassed 1 °C min^{-1} and is displayed in Table 1.

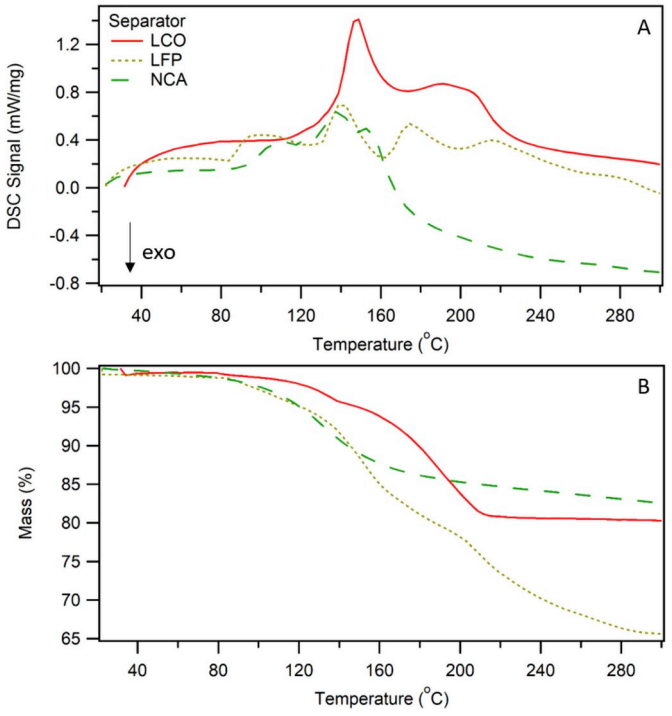


Fig. 2. Separator decomposition behavior from LCO (red, solid line), LFP (yellow, dotted line), and NCA (green, dashed line) given by the (A) DSC and (B) TGA results. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.1. Anode decomposition

All anodes experienced an exothermic transition with onset at 100–130 °C. The DSC response curves of all three anode materials as a function of SOC are shown in Fig. 3 A, B, and C (Tables S2–S4 list onset temperatures for all scenarios). In this temperature range, SEI decomposition and possible anode delithiation are the likely explanations. The SEI is comprised of both stable (inorganic LiF, Li_2CO_3, and Li_2O) and metastable (lithium alkyl carbonate and lithium semi-carbonate) components [19,35–38]. Since all anodes have metastable SEI components, the DSC shows qualitatively similar behavior. Around 100–150 °C, metastable complexes in contact with electrolyte and lithium may transform into stable SEI components, releasing small organic compounds.

Recent research suggests that the SEI is not static, but instead changes with cell voltage [39–42]. Upon charging, the SEI reversibly thickens from additional LiF, Li_2O, and other inorganic, lithium-containing components. The impact of increasing lithium (from increasing SOC) on the mass loss during metastable SEI decomposition is two-fold:

(1) a lesser amount of metastable SEI may be available for decomposition, resulting in a lower percentage of mass loss in the anode as a function of increasing SOC. This is evidenced by the TGA results in Fig. S4 A, B, and C. and
(2) the decomposition pathway may be altered by additional lithium to produce fewer gaseous products, as shown in the following equations [35]:

$$\text{original pathway}: \ (CH_2OCO_2Li)_2 \rightarrow Li_2CO_3 + C_2H_4 + CO_2 + \frac{1}{2}O_2 \quad (1)$$

$$\text{additional Li}: \ 2Li + (CH_2OCO_2Li)_2 \rightarrow 2Li_2CO_3 + C_2H_4 \quad (2)$$

By comparing DSC of a washed and unwashed anode, we show evidence of this metastable SEI contribution. A 0% SOC electrode fragment was washed with dimethyl carbonate to remove metastable SEI components and electrolyte components like $LiPF_6$, which has also been shown to trigger SEI decomposition via PF_5 attack [43]. The exothermic peak in the resulting DSC measurement (Fig. S5A) was significantly reduced together with the percentage of the mass loss, from 8.7% in the unwashed to 6.2% in the washed sample by 200 °C. This supports our assignment of the DSC anode exotherms to the decomposition process of metastable SEI components.

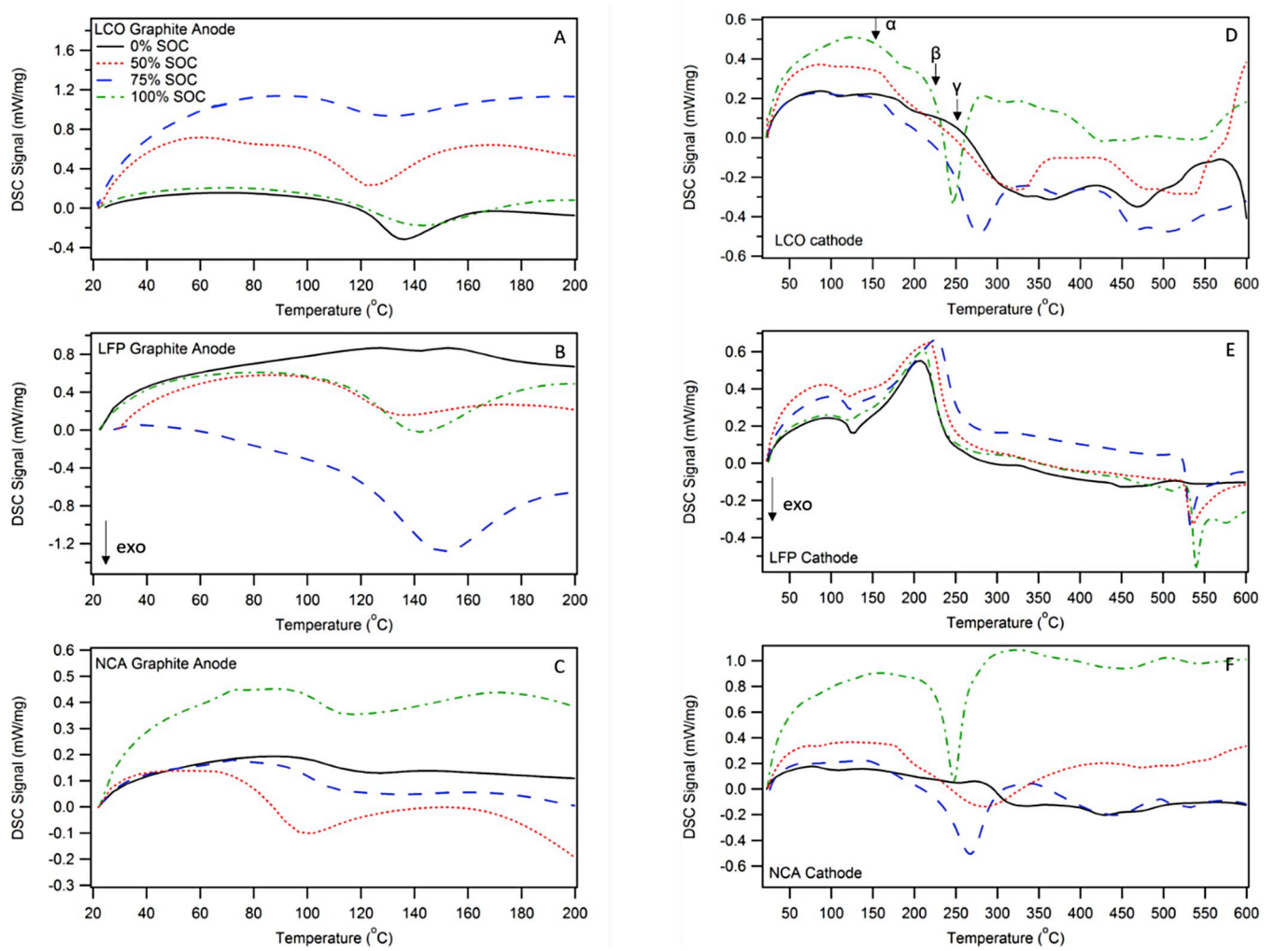


Fig. 3. DSC results from (A) LCO anode, (B) LFP anode, (C) NCA anode, (D) LCO cathode, (E) LFP cathode, and (F) NCA cathode at 0% SOC (black, solid line), 50% SOC (red, dotted line), 75% SOC (blue, dashed line), and 100% SOC (green, dash-dot line). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

XRD of LCO, LFP, and NCA anodes indicates the presence of the same phases for all chemistries at 0% and 100% SOC. Each sample was run as a film on a copper foil current collector, without any processing after it was removed from the battery. At 0% SOC, X-ray diffractograms of all three types of anodes (LCO, LFP, and NCA) are indistinguishable and all reflections can be assigned to graphite (Fig. S6A). The copper peaks observed in the diffractograms are from the current collector. Upon complete lithiation to 100% SOC, all three types of anodes again demonstrate the presence of the same phases – here, a combination of LiC_6 and LiC_{12} (Fig. S6B).

Component and whole cell thermal release can be correlated to phase changes in electrodes to help understand the multi-scale behavior in cell failure using TR-XRD. It can provide a roadmap of average structural changes through the process of thermal decomposition by selecting temperatures of interest from thermal behavior. TR-XRD measurements were conducted with all anode materials at 100% SOC. As a representative sample, the anode from an LCO cell at 100% SOC is shown in Fig. 4. With temperature increase, the degree of lithiation decreases, as evident from the transition of LiC_6 into LiC_{12} and subsequently into LiC_{18} [44]. This thermal delithiation is likely due to Li diffusion with rebalancing across the graphite matrix and/or reaction with lingering electrolyte. (This phase evolution resulting in delithiation is distinct from oxidative delithiation of graphite, shown in Fig. S7.) Anode delithiation starts around 100 °C, which corresponds to the temperature for

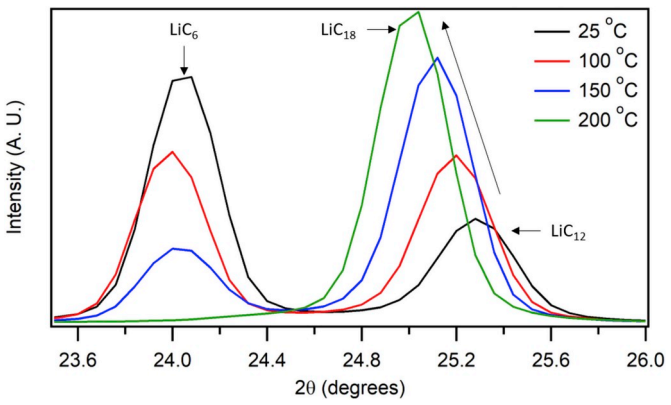


Fig. 4. Temperature-resolved XRD pattern of the 100% SOC LCO anode at 25 °C (black line), 100 °C (red line), 150 °C (blue line), and 200 °C (green line). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

SEI decomposition observed in DSC. As metastable SEI components convert into inorganic SEI, the process leaves underlying lithiated graphite unprotected, making these two decomposition processes intimately coupled. As mentioned previously, SEI/anode decomposition is

the first event in the thermal runaway cascade. As such, the LCO SEI decomposition and anode delithiation onset temperatures correspond to the ARC T_{onset} given in Table 1 of ~106 °C for all SOC. (Comparable TR-XRD results for LFP and NCA at 100% SOC are given in Fig. S8.) In the LFP anode, at 100 °C, minimal decomposition is noted; however, by 150 °C all of the LiC_6 is gone in favor of LiC_{12}/LiC_{18}. By 200 °C, the LiC_{12}/LiC_{18} has started to decompose into graphite. Similarly, by anode DSC and whole cell ARC, the LFP onset temperature is just past 100 °C (Fig. 3B and Table 1). Changes in the NCA anode were similar to LCO: by 100 °C, the LiC_6 has begun decomposing into LiC_{12}, and by 200 °C, all lithiated graphite has decomposed into LiC_{18} and an unidentified phase. The NCA anode decomposition evident by TR-XRD is consistent with the NCA anode DSC onset temperature for SEI decomposition at 100% SOC (~100 °C, Fig. 3C). However, the NCA cells exhibited an unusually high ARC T_{onset} of 133 °C (Table 1). One possible explanation is that SEI decomposition occurs shortly before the onset of anode decomposition, and it is the anode decomposition which provides more heat for the onset of the whole cell self-heating. If the NCA graphite anode possessed a surface area lower than that of the other cells, the heat release would be reduced, delaying the onset of ARC self-heating [45]. It is also plausible that the whole cell onset of the different cells varies due to the presence of different electrolyte and additive mixtures; the influence of these cocktails would not have been captured in DSC evaluations of the extracted anodes (Tables S2–4 provide a summary of onset temperatures across the different characterization techniques for all cell types and SOC.)

3.2. Cathode decomposition

The second major step in the chemical thermal runaway process is the reaction between oxygen from the decomposing cathode and the electrolyte. The onset of this reaction can be detected on the ARC by a large increase in self-heating. The exothermic cathode decomposition is responsible for the peak observed in the ARC data, $HR_{max,nominal}$ and $HR_{max,fraction}$ (Table 1). DSC heat flow plots from all three cathode materials as a function of SOC are shown in Fig. 3 D, E, and F. As before, DSC measurements were performed on the disassembled electrode materials without any treatment. From ARC measurements, LFP generates minimal thermal runaway. LFP cells with SOC below 100% do not cross the 1 °C min^{-1} threshold, though the temperatures still ultimately rise to over 400 °C. The DSC (Fig. 3E) shows a single endothermic peak centered around 200 °C for all LFP cathode materials (corresponding TGA is shown in Fig. S4E). This peak is likely related to $LiPF_6$ and/or electrolyte decomposition, as the endotherm and mass loss are consistent across SOC [46]. Such electrolyte decomposition likely occurs in LCO and NCA samples as well, although it is masked by an overlapping and overwhelming exotherm from the cathode. Previous DSC reports have suggested that LFP cathodes exhibit an exotherm at 250–300 °C [31], however, that peak is likely due to residual organic solvent on the electrode. Studies where the LFP cathode was washed indicated no peaks up to 350 °C [30,47]. In the present study, the LFP cathode TGA-DSC measurements indicate substantial decomposition around 500–550 °C, with an exothermic peak and mass loss that increase in intensity with SOC. Temperature-resolved XRD confirms this cathode behavior, as shown in previous studies and Fig. 5 [48–50]. The Li_xFePO_4 material is stable until 300 °C, but by 550 °C, the cathode has largely decomposed to $Fe_2P_2O_7$ and $Fe_7(PO_4)_6$. Due to the stability of the $LiFePO_4$ cathode, the major peak observed in Fig. 1B ARC data is likely from anode and electrolyte decomposition.

DSC of the LCO cathode in Fig. 3D reveals an initial small exothermic peak onset at ~150 °C (α), immediately followed by a larger exothermic peak (β) whose maxima and definition increases with increasing SOC. At lower SOC (0% and 50%), the peak onset (γ) and maxima are shifted to higher temperatures. Exotherm definition is lost at 50% SOC and becomes hard to definitively pull out of the baseline at 0% SOC. Upon charging (cathode delithiation), layered LCO becomes metastable and

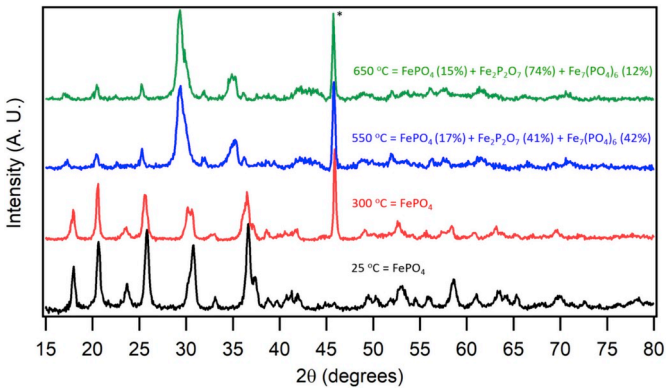


Fig. 5. Temperature-resolved XRD of 100% SOC LFP cathode. * is an impurity peak associated with the sample holder used for TR-XRD (the 25 °C sample used a different holder).

loses oxygen when subjected to temperatures above 200 °C [51]. These two exotherms match well with the two mass losses seen in Fig. S4D TGA results. Decreasing SOC results in decreasing mass loss at higher temperatures, representing a more stable cathode material. The oxygen mass loss beyond 200 °C was confirmed by temperature-resolved XRD of the 100% SOC LCO cathode (Fig. 6A). CoO_2 decomposition is expected to go as [17]:

$$Li_xCoO_2(\text{layered}) \rightarrow Co_3O_4(\text{spinel}) + O_2 \rightarrow CoO(\text{cubic}) + O_2 \quad (3)$$

As can be seen from the XRD patterns in Fig. 6A, this decomposition is not smooth, and multiple phases exist simultaneously, suggesting that there are kinetic or diffusion limitations. At 25 °C, the XRD profile is a convolution of Li_xCoO_2 and Co_3O_4 patterns. As the temperature increases, Li_xCoO_2 is gradually decomposed into increasing concentrations

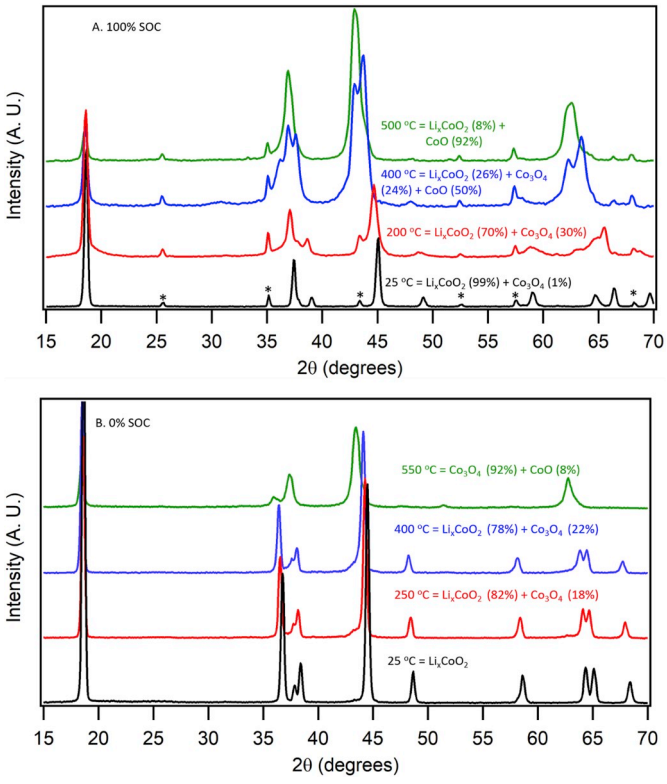


Fig. 6. Temperature-resolved XRD pattern of the (A) 100% SOC and (B) 0% SOC LCO cathode. Alumina peaks exist at all temperatures surveyed for the 100% SOC sample and are indicated with an *.

of Co_3O_4 and CoO. This results in the complex exotherms observed in the DSC (Fig. 3D) and gradual TGA mass loss above 200 °C (Fig. S4D). At 0% SOC, the cathode is far more stable, resulting in a reduced DSC exotherm and TGA mass loss. The TR-XRD pattern of the 0% SOC LCO cathode in Fig. 6B shows no major decomposition until temperatures rise beyond 400 °C, as expected from the higher stability of fully lithiated cathode material [51]. The SOC dependence evident in the cathode TR-XRD complements the trend in increasing HR_{max} with SOC observed in ARC data, Fig. 1A.

The behavior of NCA is very similar to LCO, here with a one-step exotherm observed in Fig. 3F and onset of ~150 °C (In Fig. S4F, the TGA mass loss identifies two distinct steps. The first distinct mass loss stage is followed by the second gradual loss until 550 °C. This again can be attributed to the metastable nature of the NCA cathode upon charging.) Just as in the LCO cell, lower SOC materials experience increased DSC onset temperatures and decreased exotherms and mass loss. Temperature-resolved XRD was performed on NCA at 100% SOC, and more distinct decomposition steps were noted (Fig. 7A). The decomposition pathway was expected to be the same as described for LCO above (Eq (3)) [52,53]. We fit the diffraction profile to Li_xCoO_2 (DB card # 01-076-3173) because the crystal structure does not change drastically with transition metal substitution (Co or Ni). At 25 °C, only the layered transition metal oxide exists, Li_xMO_2 (where M refers to the transition metal, either Co or Ni). At 200 °C, a majority of Li_xMO_2 has decomposed into M_3O_4. Comparing to LCO in Fig. 6A, the change in crystal structure is much more pronounced for NCA. This means that more O_2 and heat was released as more material decomposed. By 500 °C, mostly MO remains. This defined and rapid decomposition is likely responsible for the greater than three-fold increase in HR_{max}, $HR_{max,nominal}$, and $HR_{max,fraction}$ observed in NCA compared to LCO cells. In the 0% SOC NCA cathode XRD spectra, Fig. 7B, the material is largely stable through 400 °C. By 600 °C, the layered Li_xMO_2 has almost entirely decomposed into MO. Just as in the LFP and 0% SOC LCO cells, the 0% SOC NCA ARC data did not appear to include a cathode contribution.

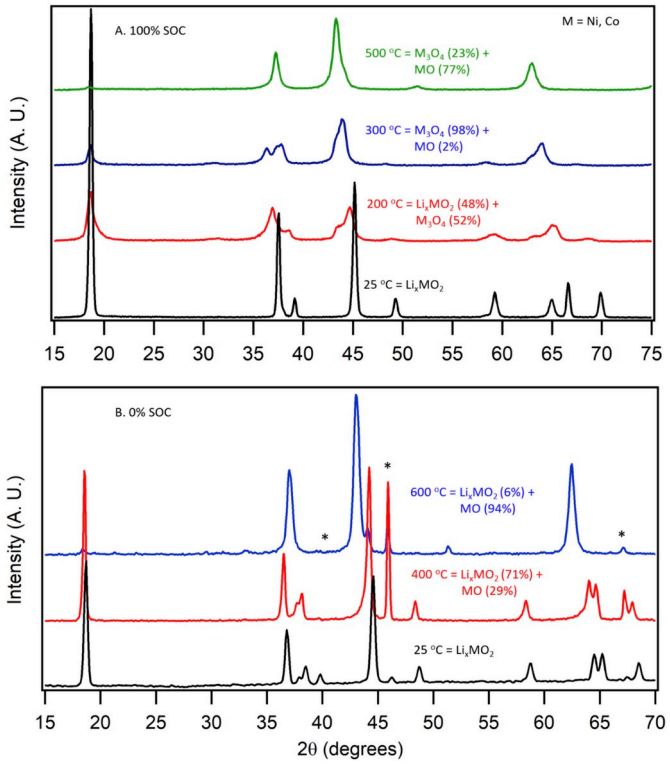


Fig. 7. Temperature-resolved XRD pattern of the (A) 100% SOC and (B) 0% SOC NCA cathode. Alumina peaks exist at all temperatures surveyed for the 0% SOC sample and are indicated with an *.

Although $LiNi_xCo_yAl_{1-x-y}O_2$ exhibits the highest rates of thermal runaway among the three cathode materials tested in this work, the fully lithiated form undergoes runaway at a rate comparable to that of LFP cells. The small exotherms observed in 0% NCA are likely tied to electrolyte decomposition.

3.3. Multi-scale comparison

Fig. 8 and Tables S2–S5 summarize the decomposition of all three chemistries across the component and whole cell scale. They demonstrate how the onset and peak temperatures of anode, cathode, and separator degradation, as measured by DSC, are related to the onset of the whole cell self-heating and thermal runaway, measured by ARC. All of the components in this study were analyzed as-received from the disassembly of commercial cells, without extensive characterization of the amount and composition of lingering electrolyte. Thus, the influence of electrode-electrolyte interactions on onset temperatures and total heat release was not fully explored in this work, though the topic has been considered in previous component level studies [54]. In this study, one possibility is that anomalous shifts in the apparent baseline of the DSC data may be attributable to varying amounts of electrolyte in the samples. Since no additional electrolyte was added to the electrodes, as is sometimes done in DSC studies, the peaks may also be smaller in magnitude than those observed in previous studies.

A key objective of the multi-scale comparison is to understand the self-consistency of different thermal characterization techniques, particularly the ability of component level characterization to predict whole cell behavior. One of the outcomes of this study was the recognition that while ARC protocols for whole cells are fairly standardized (for example, the onset of self-heating is typically defined as 0.02 °C min^{-1}), the lack of standardization for component-level DSC and TR-XRD protocols makes comparisons among different studies difficult. One critical variable is heating rate, as the finite thermal conductivity of the sample medium will affect the apparent onset temperatures of chemical processes in this sample. For example, the faster the temperature rises in ARC, DSC, or TR-XRD, the higher will be the perceived onset temperature for a reaction or phase change [21]. These are not equilibrium processes. In this study, all of the DSC analysis was initially carried out at a scan rate of 15 °C min^{-1}, which is within the range commonly used in the battery component abuse literature. To determine the heating rate effect on onset temperatures, the measurement was repeated at 3 °C min^{-1} for the 100% LCO and NCA cathodes (Fig. S9). At 3 °C min^{-1}, the LCO cathode showed an onset temperature of 208 °C, compared to 229 °C at 15 °C min^{-1}, and the NCA cathode had an onset of 218 °C, compared to 230 °C at 15 °C min^{-1}. Thus, the real equilibrium value of component-level onset temperature is likely moderately lower for all of the points indicated in Fig. 8. However, this result would not qualitatively affect the presented discussion as the relative positions of the data points are still the same across different measurements (for example, cathode DSC onset temperatures are still significantly above the ARC onset of thermal runaway). With this in mind, the following sections lay out the proposed degradation pathway for each chemistry, relying on all thermal characterization techniques.

3.3.1. LCO decomposition

When charged, the LCO cells appear to fail as a result of anode and separator decomposition. For these cells, the whole cell ARC self-heating onset temperature was close to the anode DSC exotherm onset (about 110 °C). The latter coincided with phase transition of LiC_6 to LiC_{12} by 100 °C, as evidenced by TR-XRD of the anode component. The exothermic decomposition of the SEI and anode fueled the endothermic degradation of the separator (peak at 148 °C according to component level DSC). A combination of a soft short circuit from gradual separator decomposition and a developing thermochemical reaction enabled whole cell thermal runaway, with onset temperatures ranging from 168 to 198 °C (from 100% to 50% SOC). In all charged states, the ARC onset

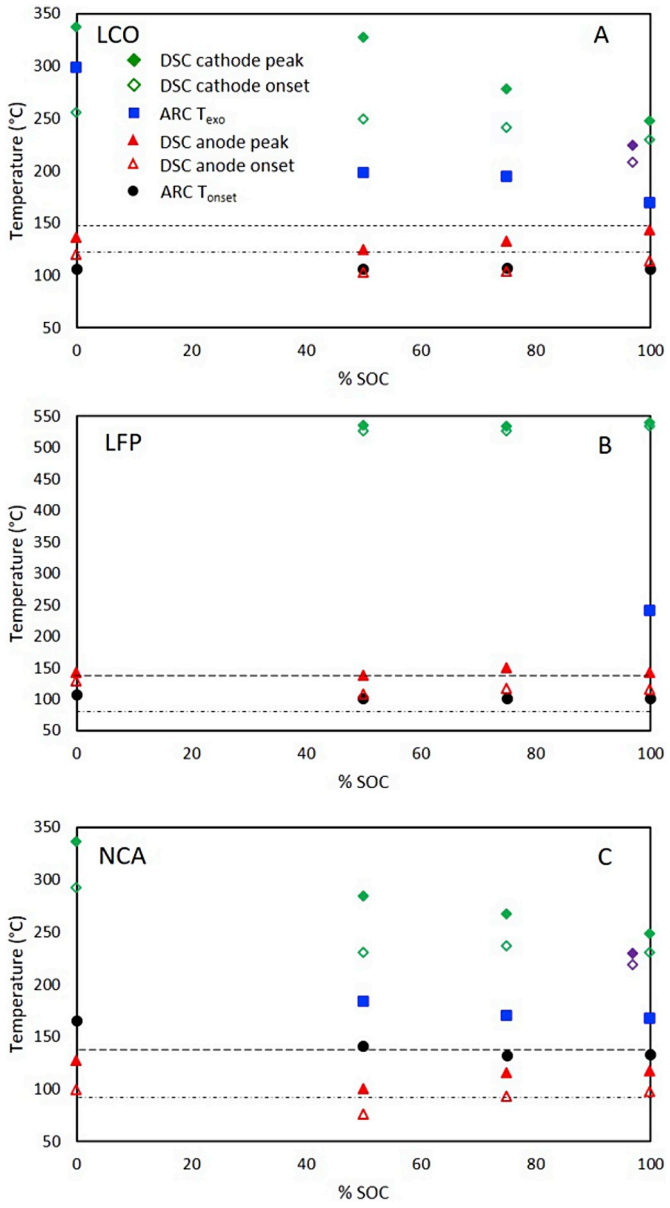


Fig. 8. Comparison of component and whole cell degradation onset and peak temperatures for (A) LCO, (B) LFP, and (C) NCA cells. The dash-dot line is the DSC onset of the endotherm for separator breakdown and the dashed line is the peak. Green diamonds represent DSC cathode data and red triangles DSC anode data. Outlined points represent onset temperatures and filled represent peak temperatures. Black dots represent temperature onset in ARC while blue squares represent T_{exo} in ARC. The filled purple diamond is the DSC cathode peak and the empty purple diamond the cathode onset for 100% SOC at a scan rate of 3 °C min^{-1} (offset on the x-axis for easier viewing). The rest of the DSC data was collected at 15 °C min^{-1}. If a data point is not given, the designated thermal event did not occur. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

of whole cell runaway preceded both the DSC onset of the major cathode exotherm (229–249 °C, from 100% to 50% SOC) and the initiation of a TR-XRD phase change from Li_xCoO_2 to Co_3O_4 (just below 200 °C). These results suggest that thermal runaway is initiated primarily by the anode and separator breakdown. Therefore, we posit that improving cathode component stability alone is insufficient to lower cell runaway onset temperatures, though it can lower the overall heat release.

3.3.2. NCA decomposition

Similar to LCO cells, the failure of NCA cells occurs before the cathode degradation. The NCA anode showed DSC onset temperatures slightly lower than those of the LCO anode (consistently < 100 °C). However, while the LCO cell showed close agreement between anode degradation and the whole cell ARC self-heating onset, in the NCA cell, the whole cell onset was consistently at least 30 °C higher than the onset of DSC anode exotherms and the TR-XRD phase transition of LiC_6 to LiC_{12}. This gap between the start of the anode component and the whole cell degradation suggests an unevaluated thermal role of the electrolyte, which would not have been adequately captured in DSC evaluation of anodes extracted from the cells. Previous reports of ARC on various commercial 18650 NCA cells at 100% SOC have indicated self-heating onset temperatures ranging from 51 °C to 100 °C (133 °C in this study) [30,55]. This degradation enables separator breakdown, evidenced by the substantial DSC endotherm at 140 °C. As with the LCO cell, the onset of thermal runaway in the ARC for a charged cell (166–182 °C, from 100% to 50% SOC) preceded both the DSC cathode exotherm, ~230 °C, and significant TR-XRD phases changes (Li_xTMO_2 to TM_3O_4).

3.3.3. LFP decomposition

In the LFP cells, the anode DSC exotherm onset and whole cell ARC self-heating onset were consistent with that of LCO cells and showed no SOC dependence. Again, the SEI and anode decomposition was followed by separator breakdown at 141 °C. However, in this case, separator breakdown did not cause whole cell thermal runaway due to the inherent stability of the cathode material. Cathode component DSC showed exotherm onset temperatures of ~530 °C and TR-XRD phase changes (from $FePO_4$ to $Fe_2P_2O_7$ and $Fe_7(PO_4)_6$) were apparent between 300 and 550 °C. Only the 100% SOC LFP cell exhibited a heating rate above 1 °C min^{-1} (by 241 °C), but the maximum heating rate was still only 4.5 °C min^{-1}, orders of magnitude smaller than LCO and NCA.

The different degradation processes showed a mixed dependence on SOC. Except for the NCA cells, neither the DSC anode onset nor the ARC whole cell self-heating onset showed SOC dependence. However, among all cells, the ARC onset temperature of whole cell thermal runaway decreased and the overall heat release increased with increasing SOC. Similarly, by DSC, all of the cathodes showed increasing exotherm magnitude with increasing SOC. The fact that the SOC has little influence on whole cell and anode onset temperatures but does influence the start of runaway and overall heat release is relevant to addressing the safety concerns of LIB systems.

4. Conclusions

We have investigated the multi-scale thermal stability of Li-ion battery failure at the component material level using TGA/DSC and temperature-resolved XRD. This component decomposition is compared to whole cell ARC failure. Component data confirms that the exothermic decomposition of the SEI and anode is responsible for whole cell self-heating onset temperatures. SEI decomposition is observed in DSC and TGA data. TR-XRD revealed anode delithiation begins at 100 °C when charged to 100% SOC, and is similar for all anodes. Energy from anode decomposition then catalyzes separator melting. The fact that the ARC onset of whole cell thermal runaway was consistently at least 30 °C above the DSC endotherm peak for the separator suggests that runaway is fueled by a developing short circuit from the decomposing separator coupled with thermochemical reactions, rather than a hard short. The high runaway rates present for LCO and NCA cells exceed the capabilities of the ARC to provide an adiabatic environment and the data should be considered qualitative. However, a trend still exists in HR_{max}, $HR_{max,nominal}$, and $HR_{max,fraction}$ when comparing LCO and NCA. According to DSC and TR-XRD, the NCA cathode decomposes suddenly, releasing oxygen and heat rapidly upon heating, whereas the LCO cathode decomposes stepwise. This correlates to the three-fold or greater HR_{max}, $HR_{max,nominal}$, and $HR_{max,fraction}$ observed in ARC for NCA cells compared to LCO cells. The LFP cells had the lowest HR_{max} and $HR_{max,nominal}$ of all cells tested. This can be attributed to the stability of

the $LiFePO_4$ cathode, which did not decompose until after 500 °C during DSC evaluation. This detailed and comparable thermal stability information on various cell chemistries at different SOCs is valuable when engineering future failure mitigation strategies or inherently safe cells. Understanding the cascade steps from thermal runaway onset to maximum heat release is critical when rationally approaching a safer cell or system design.

Author contributions

H.M.B. and S.R.F. designed this work. H.M.B. and S.I. conducted DSC and TR-XRD. J.L., L.T., and J.L. carried out cell disassembly and ARC. H. M.B., Y.P., and S.R.F. wrote the paper. All authors participated in the analysis of the experimental data and discussion of the results, as well as the preparation of the paper. All authors have given approval to the final version of the manuscript.

Declarations of interest

None.

Acknowledgements

This work was supported by the US Department of Energy Office of Electricity, Energy Storage Program. The authors wish to thank Dr. Imre Gyuk for his support of research advancing safety in stationary energy storage. We are grateful for fruitful discussions with Randy Shurtz and John Hewson. This work was performed, in part, at the Center for Integrated Nanotechnologies, an Office of Science User Facility operated for the U.S. Department of Energy (DOE) Office of Science. Sandia National Laboratories is a multi-mission laboratory managed and operated by National Technology and Engineering Solutions of Sandia, LLC., a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA-0003525. This paper describes objective technical results and analysis. Any subjective views or opinions that might be expressed in the paper do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2019.226777.

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